

Ayush Satyam

☎ +91 9596490722  github.com/ayushsatyam146  linkedin.com/in/ayushsatyam146 ✉ ayushsatyam146@gmail.com

EXPERIENCE

Red hat

Bengaluru, India

Systems Machine Learning Engineer

Nov 2025 - Present

- Contributed to upstream **PyTorch compiler infrastructure** as a member of the **TorchDynamo upstream working group**, collaborating with Meta and PyTorch maintainers on compiler architecture, refactoring, and `torch.compile` performance.
- Worked across the **PyTorch → FX → TorchDynamo → CUDA/vLLM** stack, contributing to compiler internals, graph-break reduction, CUDA Graph execution, and inference performance.
- Authored and drove an **accepted RFC** introducing native **ContextVar support** for `torch.compile`, addressing graph breaks across **100+ Hugging Face models** and collaborating with PyTorch, Hugging Face, and vLLM maintainers.

Software Engineer - 2

Feb 2025 - Nov 2025

- Independently delivered** the Buildpacks build strategy for OpenShift Builds, **onboarding 30% new customers** and accelerating native image adoption.
- Led core design** initiatives for multi-architecture (amd64, arm64, s390x) builds and architected the OLM v1 migration, alongside **upstream OSS** features like shallow cloning, feature gating, and test suite labeling.
- Architected **ClusterBuildStrategies for CUDA and PyTorch-based workloads**, enabling optimized container builds for ML infrastructure.

Software Engineer - 1

Jul 2023 - Feb 2025

- Delivered** the OpenShift Builds GA release, **enabling 50+ enterprise customers** with seamless in-cluster image building.
- Automated** CVE detection, triage, and patching workflows, reducing security rollout times **from 1 week to near zero**.
- Mentored 10 university outreach students**, strengthening platform security and fostering **open-source contributions**.

Software Engineer intern

Nov 2022 - May 2023

- Slashed SPA deployment times by 10×** via bundle splitting and caching, boosting productivity and **reducing infra costs**.
- Cut gateway response times from 3s to 200ms** by migrating One-platform services to the Backstage internal developer portal.

Google Summer of Code (GSoC), Python Software Foundation

Remote

Student Developer

May 2022 - Oct 2022

- Migrated** EOS Icons to **Next.js and TypeScript**, improving performance, scalability, and SEO across **5000+ icons**.
- Established Jest, React Testing Library, and Cypress with CI/CD, achieving **90%+ test coverage**.

OPEN SOURCE CONTRIBUTIONS

PyTorch: (contributions)

- Contributed to the **TorchDynamo `tp_slot` refactor** alongside PyTorch maintainers, migrating `tp_repr` and `tp_str` implementations and improving compiler infrastructure maintainability.
- Represented Red Hat across the PyTorch and AI ecosystem, delivering technical talks and organizing community events across **India and Singapore**, including engagements at **IISc and IIT Bombay** on PyTorch, `torch.compile`, vLLM, and AI systems performance.

vLLM: (contributions)

- Implemented **CUDA Graph dispatch optimizations** improving cascade attention performance and KV-cache utilization.
- Resolved distributed inference reliability and correctness issues, fixing **multi-rank worker deadlocks** through RPC-based consensus and correcting **KV-cache** behavior.
- Presented a talk** on vLLM architecture and inference systems to a group of **PhDs and industry experts** at a South Korean AI inference lab .

DiceDB: (contributions)

- Core contributor to a high-performance database, spearheading critical backend enhancements:
- Engineered **Write-Ahead Logging, checkpointing, and multi-cluster replication** for a high-performance distributed database.
- Developed a **thick SDK** reducing response latency by **87.5%**, and optimized command execution through integration with a custom execution engine.

PROJECTS

MegaBake

Github 

- Built a fully automated compiler that transforms **PyTorch FX graphs into persistent CUDA megakernels**.
- Designed a compilation pipeline to **fuse GPU workloads**, reducing kernel launch overhead for inference.
- Integrated compiler-generated kernels into the **PyTorch inference workflow** for efficient repeated execution.

TECHNICAL SKILLS

- ML Systems:** PyTorch, `torch.compile`, TorchDynamo, PyTorch FX, CUDA, CUDA Graphs, vLLM, KV Caching
- Programming:** C/C++, Golang, Python, TypeScript, SQL, $\LaTeX$, Linux/Unix.
- Cloud / Infrastructure:** GCP, RHEL, OpenShift, Kubernetes, Tekton, ArgoCD, Kafka, Jenkins, Redis, MySQL, Postgres.

EDUCATION

Shri Mata Vaishno Devi University

B.Tech in Computer Science and Engineering

May 2019 - May 2023

GPA: 8.59 / 10.0